



Intelligent Transportation Systems and Traffic Operations

Time-Space Diagram

Instructor

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Review

❑ From last lecture to today

- Mobility and Accessibility
 - ✓ Transportation connects people and goods to opportunities
- System-level Challenges
 - ✓ Congestion, crashes, energy use and emissions
- Smarter Transportation Systems
 - ✓ Transportation Engineering, ITS, C-ITS



Today

To manage traffic intelligently, we need **a tool** to represent and analyze vehicle movements over time and space.

Time-Space Diagram

Overview

□ Why do we use a time-space diagram?

- Transportation users and vehicles usually move along defined paths or channels.
- Examples include cars and buses on roads, pedestrians on sidewalks, trains on tracks, ships along routes, and aircraft along flight paths.
- A time-space diagram represents this movement by plotting position against time, allowing us to analyze speed, stops, queues, and interactions.

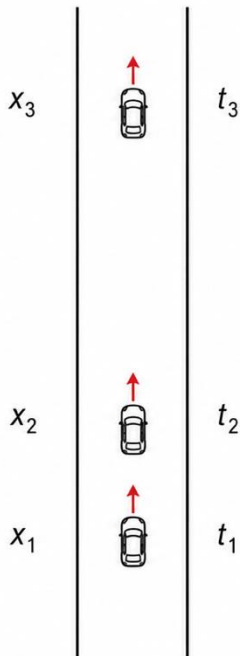
Time-Space Diagram

Single Vehicle Trajectory

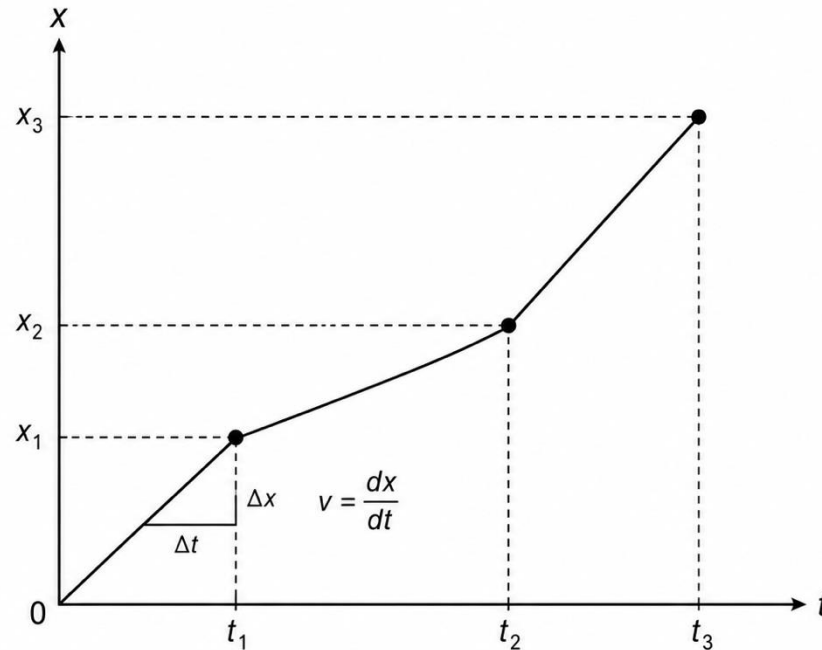
□ Time-space diagram

- Trajectories: graphical representation of $x(t)$ in (t, x) plane

Road Segment



Time-Space Diagram

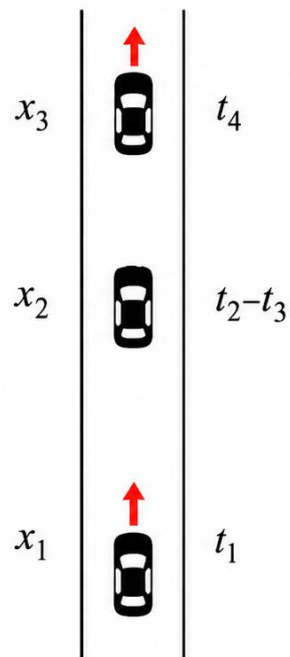


Time-Space Diagram

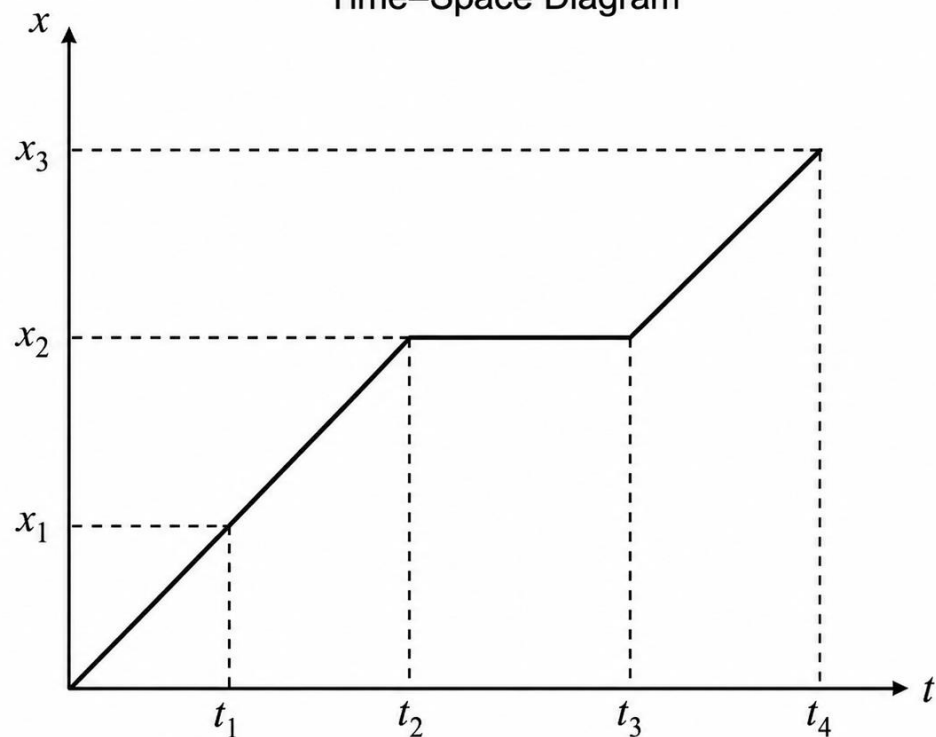
Single Vehicle Trajectory

□ Time-space diagram

Road Segment



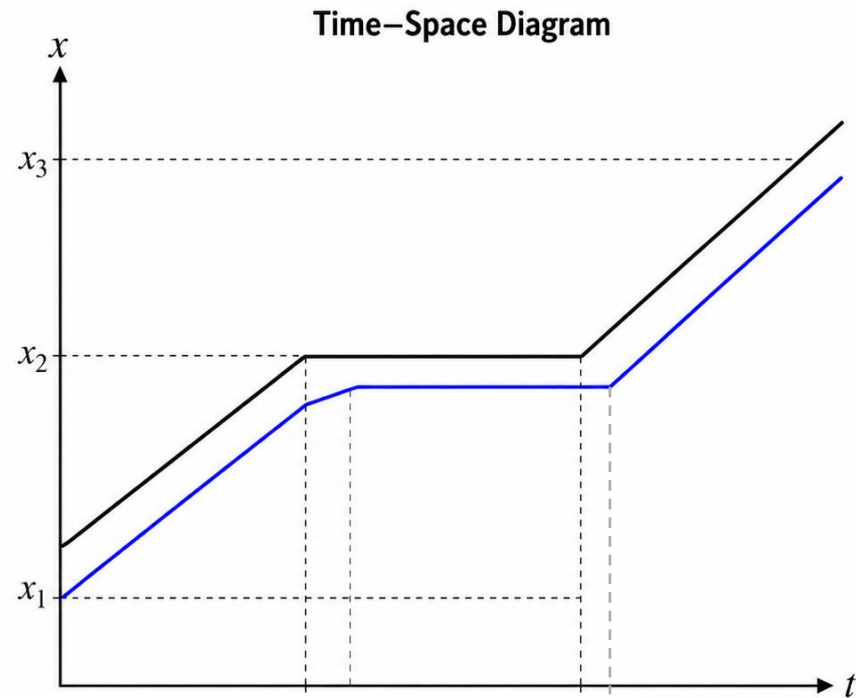
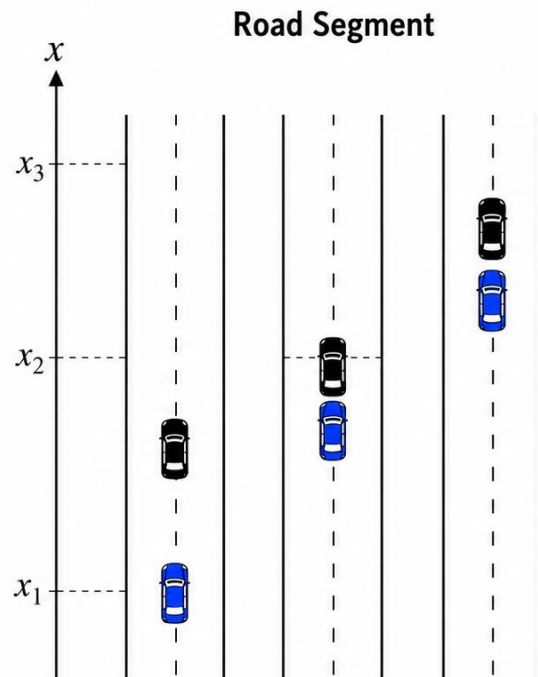
Time-Space Diagram



Time-Space Diagram

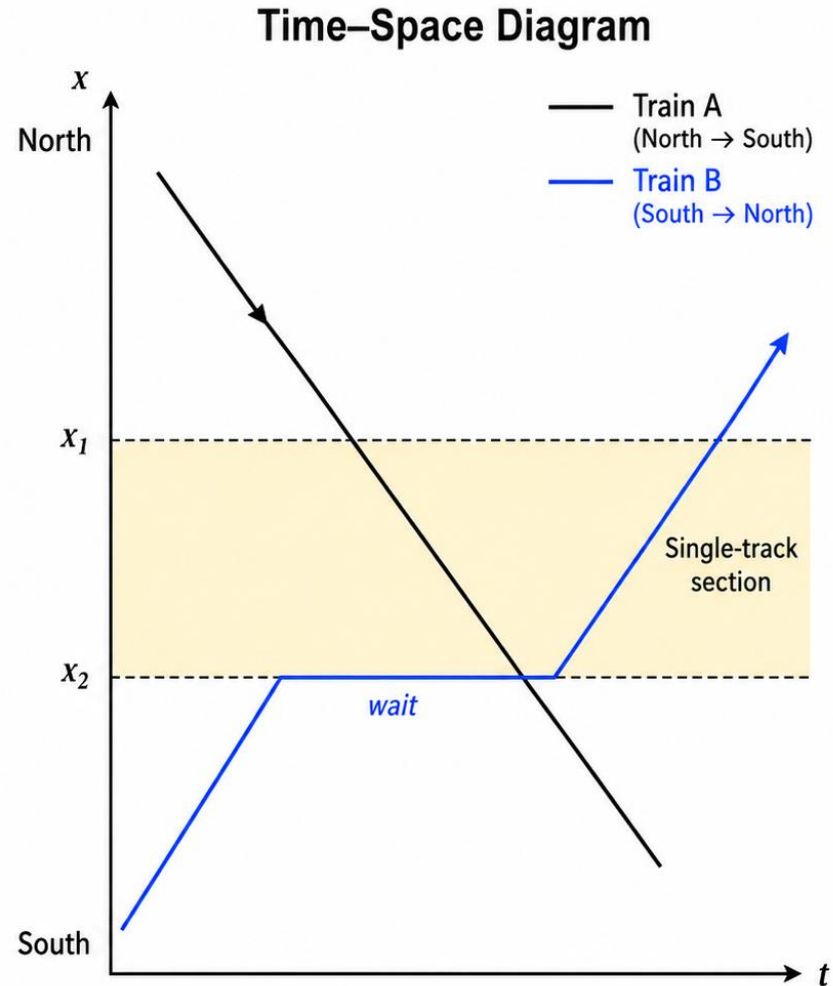
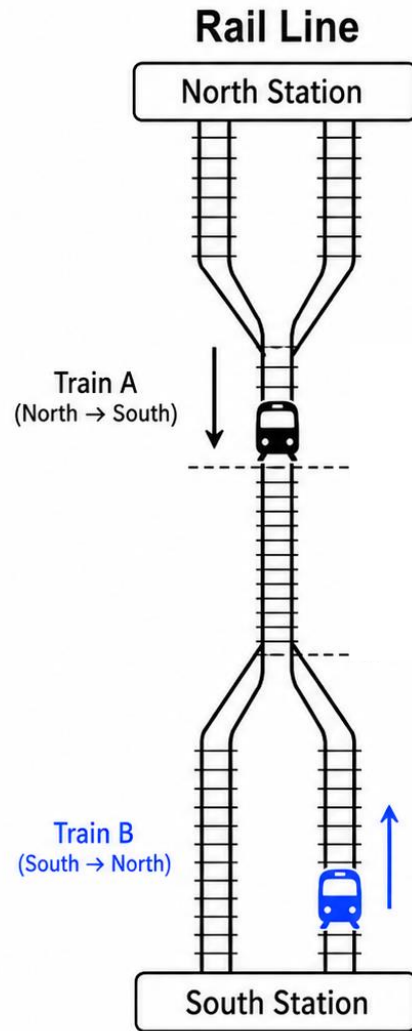
Multiple Vehicle Trajectories

□ Time-space diagram



Time-Space Diagram

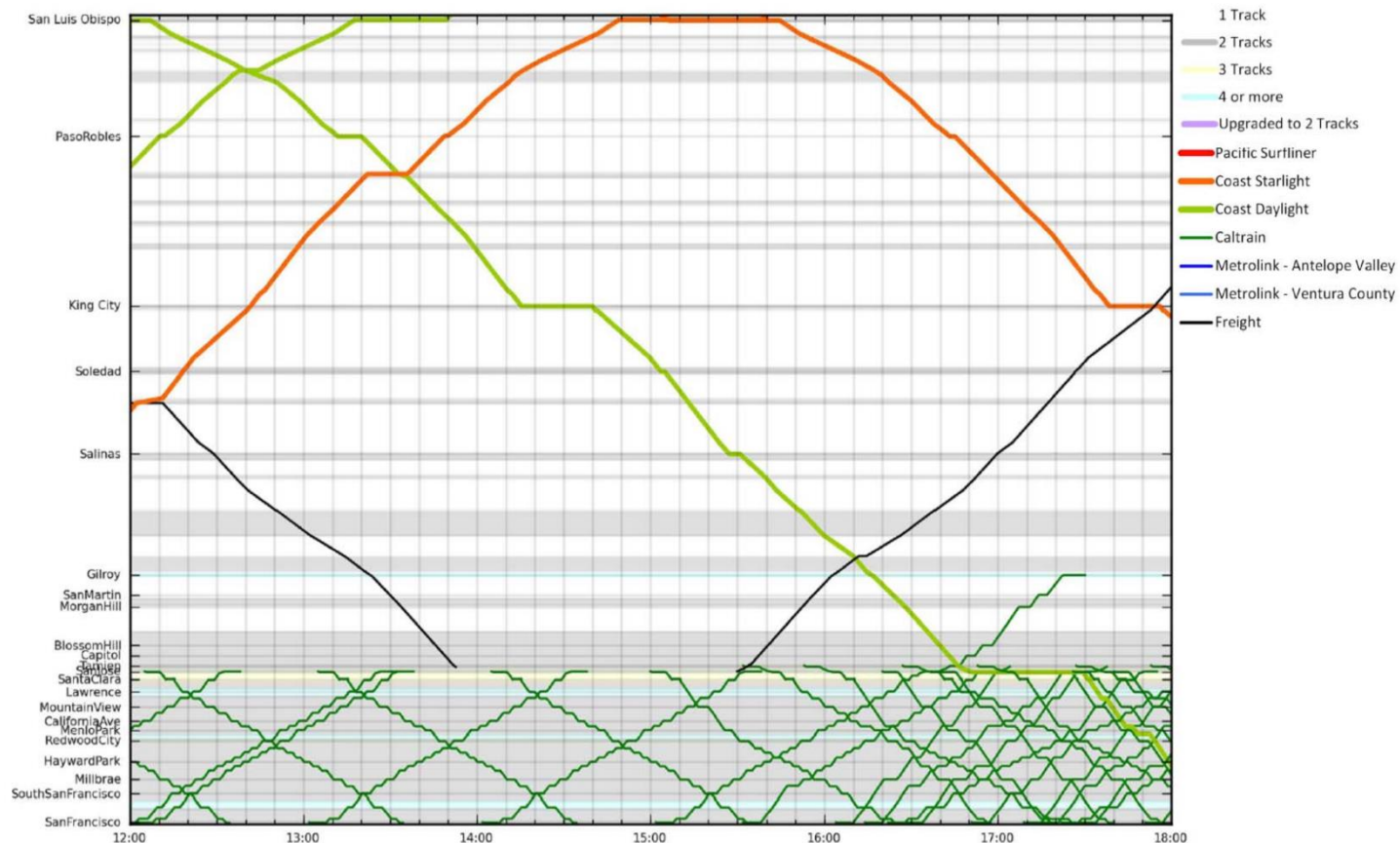
Train Operation



Time-Space Diagram

Train Operation

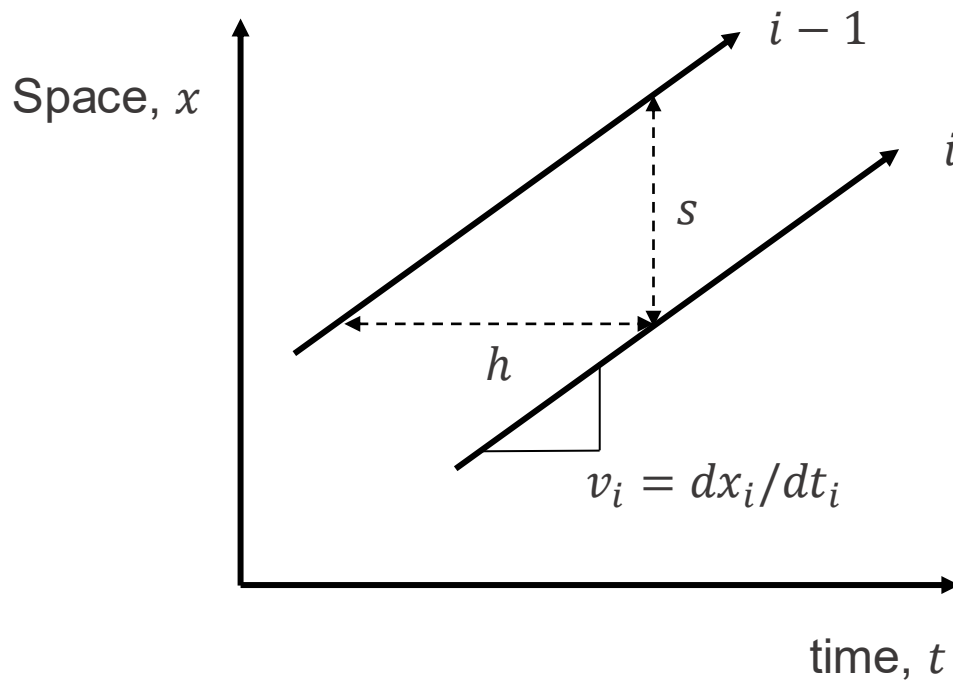
Real-World Application: Coast Corridor, California



Time-Space Diagram

Spacing and Headway

□ Traffic stream properties



s = spacing;

the distance between two vehicles

h = headway;

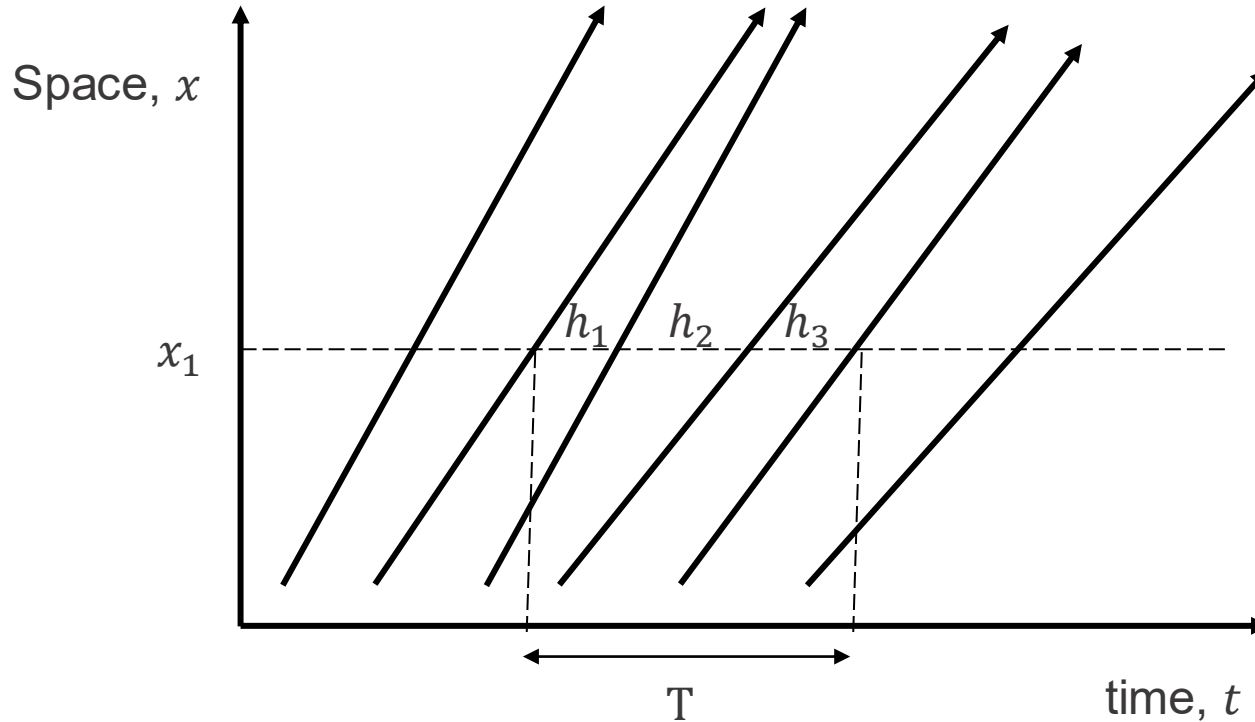
the time between two vehicles arrivals

v_i = speed of vehicle i

Time-Space Diagram

Spacing and Headway

□ Traffic flow, q



n : the number of vehicles closing location x_1 during period T

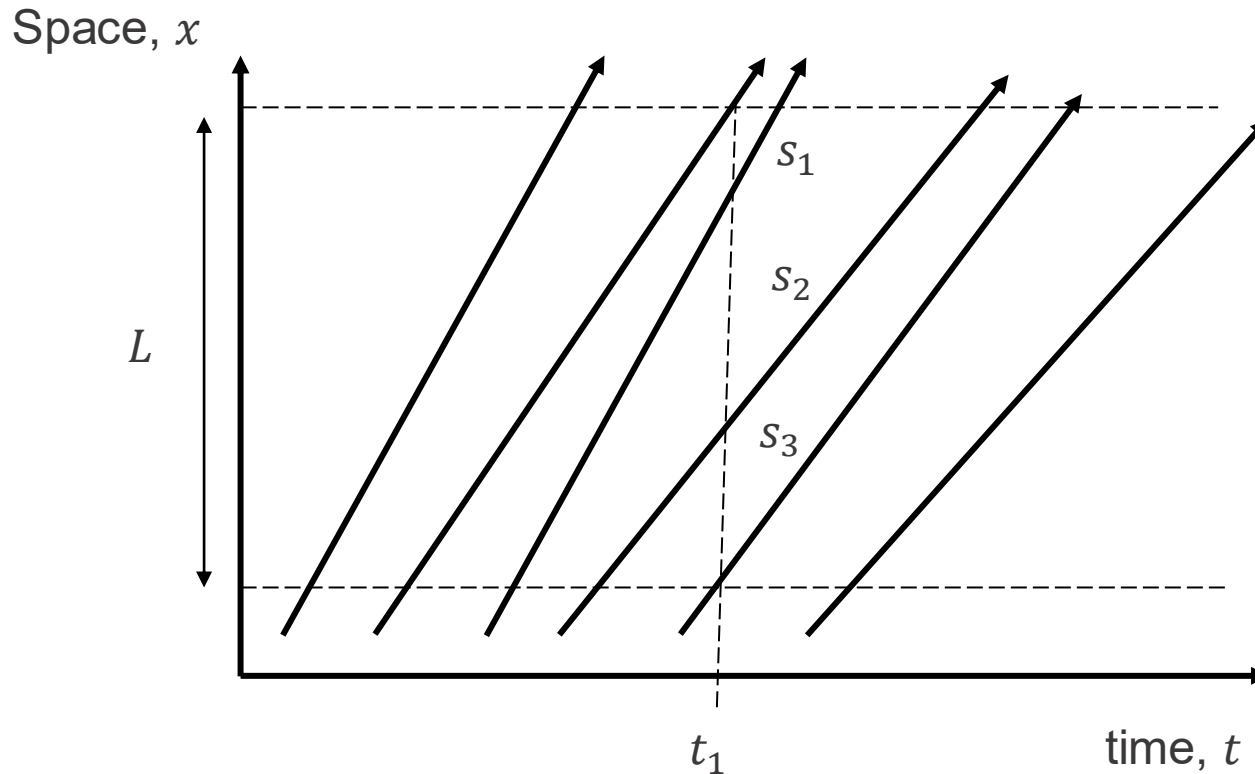
$$q = \frac{n}{T} \approx \frac{n}{\sum_{i=1}^n h_i} = \frac{1}{\frac{1}{n} \sum_{i=1}^n h_i}$$

$$q = \frac{1}{\bar{h}}$$

Time-Space Diagram

Spacing and Headway

□ Traffic density, k



m : the number of vehicles
in segment L at instant t_1

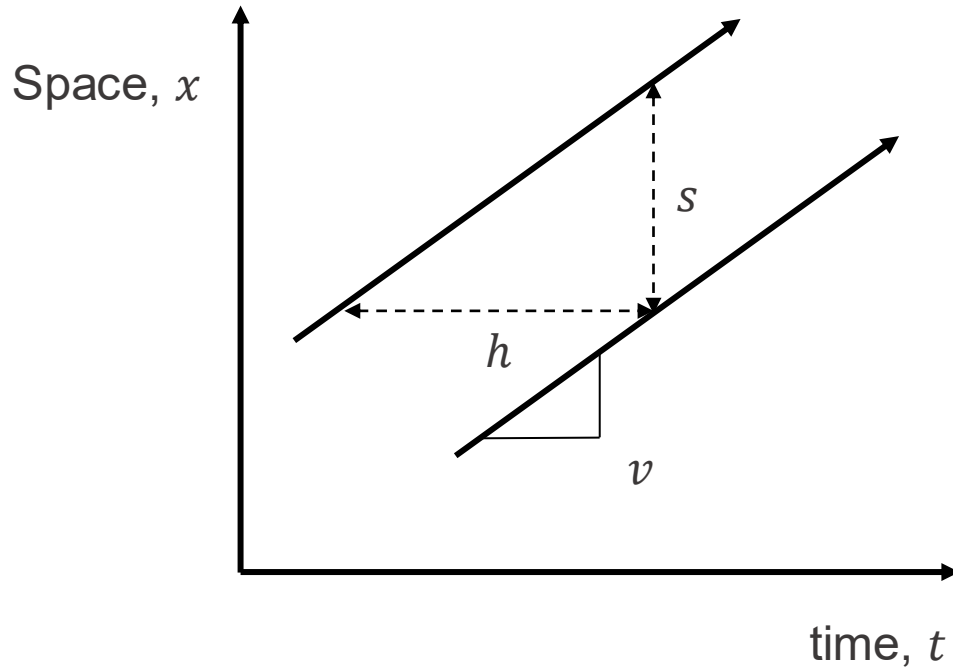
$$k = \frac{m}{L} \approx \frac{m}{\sum_{i=1}^m s_i} = \frac{1}{\frac{1}{m} \sum_{i=1}^m s_i}$$

$$k = \frac{1}{\bar{s}}$$

Time-Space Diagram

Spacing and Headway

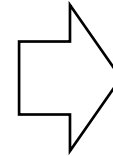
□ Traffic stream properties



$$q = \frac{1}{h}$$

$$k = \frac{1}{s}$$

$$v = \frac{s}{h}$$



$$v = \frac{q}{k}$$

$$q = kv$$

Time-Space Diagram

Spacing and Headway

$$v = \frac{q}{k} \quad q = kv$$

□ Sample Problem: Traffic Flow and Density

➤ A roadside detector on a multilane highway measures the following traffic:

- ✓ Passenger cars: 1,800 cars/hr, average speed 60 mph
- ✓ Trucks: 300 trucks/hr, average speed 30 mph

➤ Question 1

✓ Passenger-car density $k_c = \frac{q_c}{v_c} = \frac{1800}{60} = 30 \text{ cars/mi}$

✓ Truck density $k_t = \frac{q_t}{v_t} = \frac{300}{30} = 10 \text{ trucks/mi}$

✓ Total traffic density $k = k_c + k_t = 40 \text{ veh/mi}$

Time-Space Diagram

Spacing and Headway

$$v = \frac{q}{k} \quad q = kv$$

□ Sample Problem-Continued

- At a downstream location, the multilane highway narrows to a single lane.
- Cars and trucks enter the one-lane section without delay. Within the section, faster passenger cars catch up with slower trucks and form queues behind them.
- Question 2
 - ✓ Determine the average number of passenger cars queued behind each truck in the one-lane section.

$$\frac{q_c}{q_t} = \frac{1800}{300} = 6 \quad \longrightarrow \quad \text{6 cars per truck, on average}$$

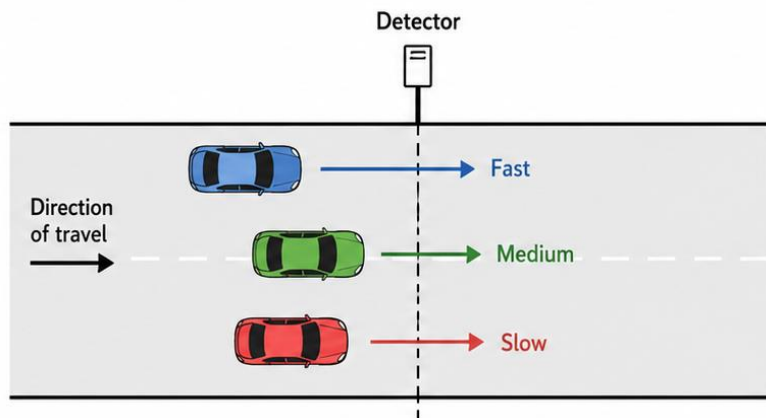
Time-Space Diagram

Time- and Space-Mean Attributes

❑ Does the average speed depend on how vehicles are observed?

Point measurement

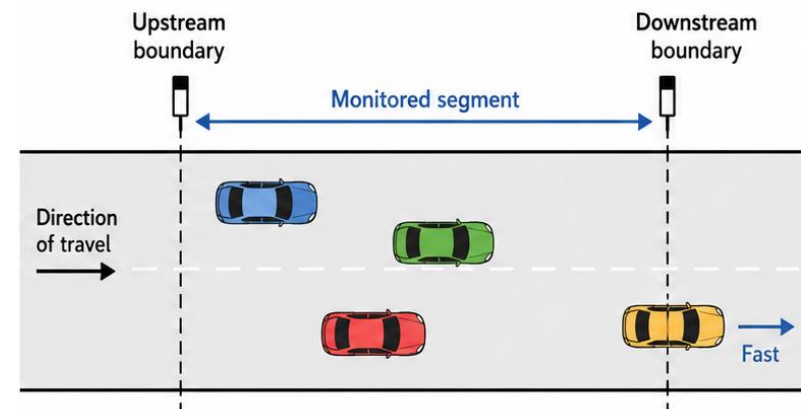
Observe vehicles as they pass on location



Time-mean speed

Segment measurement

Observe vehicles occupying a road segment



Space-mean speed

Time-Space Diagram

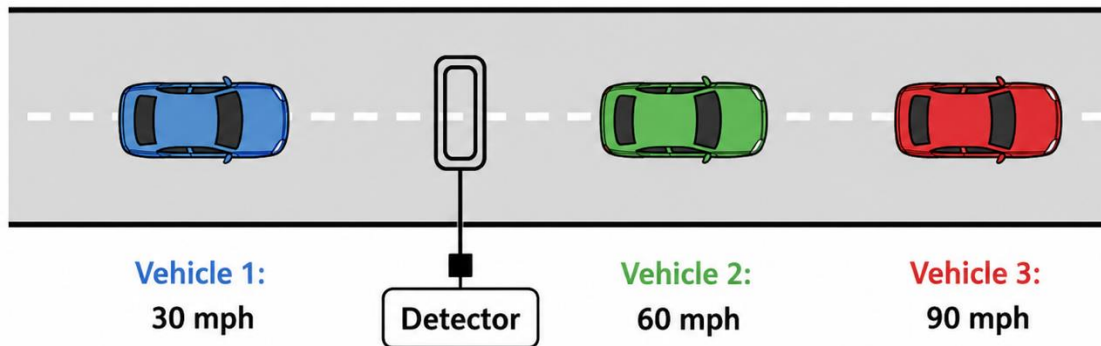
Time- and Space-Mean Attributes

□ Time-mean speed: average at a point

- The arithmetic mean of vehicle speeds observed at a fixed location over time.

$$\bar{v}_t = \frac{1}{n} \sum_{i=1}^n v_i$$

- Example



$$\bar{v}_t = \frac{30 + 60 + 90}{3} = 60 \text{ mph}$$

Time-Space Diagram

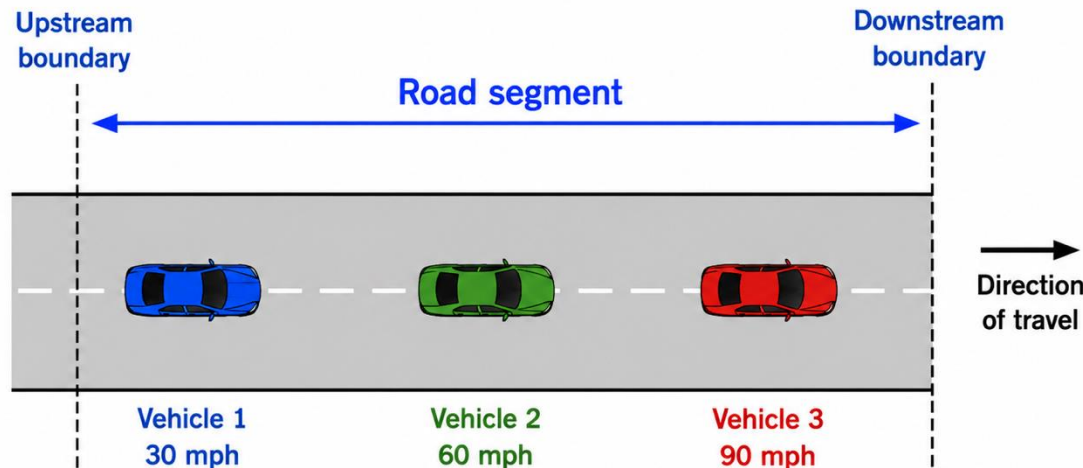
Time- and Space-Mean Attributes

□ Space-mean speed: average at a point

- The average speed of vehicles traveling over a specified road segment.

$$\bar{v}_s = \frac{\text{total distance traveled}}{\text{total travel time}} = \frac{nL}{\sum_{i=1}^n L/v_i} = \frac{n}{\sum_{i=1}^n 1/v_i}$$

➤ Example



$$\bar{v}_s = \frac{3}{\frac{1}{30} + \frac{1}{60} + \frac{1}{90}} = 49.1 \text{ mph}$$

Time-Space Diagram

Time- and Space-Mean Attributes

□ Why Is Space-Mean Speed Lower?

$$\bar{v}_s = 49.1 \text{ mph} < \bar{v}_t = 60 \text{ mph}$$



- **Slower vehicles** remain in the segment longer, so they receive **greater weight** in space-mean speed.

Time-Space Diagram

Time- and Space-Mean Attributes

□ Time-mean speed vs. Space-mean speed

	Time-Mean Speed	Space-Mean Speed
Observation	Fixed point	Road segment
Averaging method	Arithmetic mean	Harmonic mean
Weighting	Equal per vehicle	Greater weight to slow vehicles
Typical data	Spot-speed detector	Travel time / trajectory data
Main interpretation	Speeds passing a point	Traffic stream over space